MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE DOD-008 1/

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Passive TR-limiter, crystal protector, frequency range 8,750 to 9,650 MHz

ABSOLUTE RATINGS:

Parameter:	Incident power	tp	prr	TA	Alt
Unit:	kw	μs	pps	°C	ft
Maximum:	100	2.0	525	71	10,000
Minimum:	---	0.5	---	-54	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Weight:	10 ounces (max)
Flanges:	Mates with UG-136B/U	Finish:	See figure 1
Marking:	See figure 1		

TEST CONDITIONS:

Parameter:	Incident power	tp	prr	Pressurization	TA	F
Unit:	kw	μs	pps	psia	°C	MHz
Tolerance:	±10%	±0.10	---	±1.0	max	±1%
Test condition 1:	6	0.5 or 2.0	525	30	+71	F3
Test condition 2:	100	0.5 or 2.0	525	30	+71	F3

FREQUENCY		
F	MHz	±%
1	8,750	0.1
2	8,975	0.1
3	9,200	0.1
4	9,425	0.1
5	9,650	0.1

GENERAL:

Qualification - Not required

1/ Replaces type MD-80X33.

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1	-					
4473	Low-level VSWR	3	-	F1, F2, F3, F4, F5	σ	---	1.3	---
4416	Insertion loss	3	-	F1, F2, F3, F4, F5	Li	---	0.8	dB
4452	Flat-leakage power	-	1		pf	---	30	mw
4452	Spike-leakage energy	-	1		Ws	---	0.05	erg
4496	Firing power	-	1		Po/Du	---	150	mw
1027	Temperature cycling (nonoperating)	-	-	1 cycle; TA = -54° to +71° C	---	---	---	---
	Quality conformance inspection, part 2							
4471	Recovery time	2	2		t	---	2	μs
4003	Pressurizing	-	-	30 psia	---	---	---	---
---	Linearity	4	1		---	---	---	---
	Quality conformance inspection, part 3							
---	Life test	-	2		t	2,000	---	hrs
---	Life-test end points:							
4471	Recovery time	2	2		t	---	4	μs
4452	Flat-leakage power	-	1		pf	---	50	mw
4452	Spike-leakage energy	-	1		Ws	---	0.1	erg
4416	Insertion loss	3	-	F1, F2, F3, F4, F5	Li	---	1.0	dB
	Periodic-check tests							
4021	Degradation due to vibration	-	-		---	---	---	---
1042	Shock, specified pulse	-	-	40 G	---	---	---	---
1027	Temperature cycling life test	-	-	10 cycles (min)	---	---	---	---
4471	Recovery time	2	2	F1, F3, and F5	t	---	2	μs
1011	Humidity	-	-	0 to 95 percent	---	---	---	---

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NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
2. The recovery time shall be measured at the point where the signal level has recovered to within 0.5 dB of the full-signal level.
3. A swept-frequency method may be used for this test.
4. The compression shall be 1 dB above +3 dBm, or 1 dB above the insertion loss specified under quality conformance inspection, part 1.

Custodian:
Air Force - 11

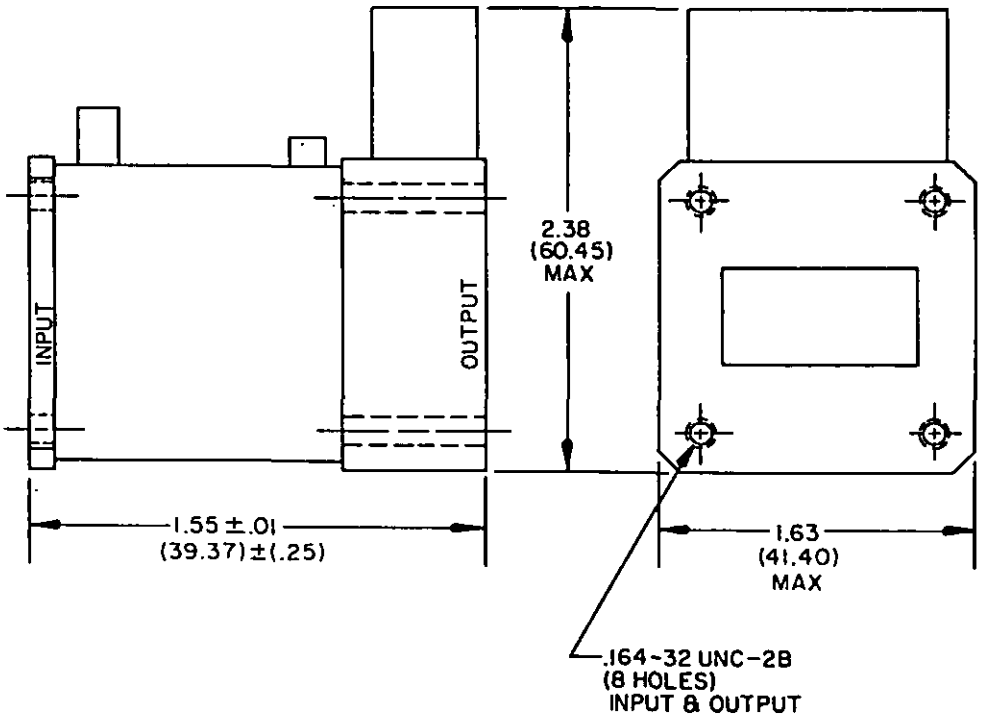
Review activities:
Air Force - 11, 17, 80

Preparing activity: Air Force - 11

Agent: DSA - ES

(Project 5960-F945)

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NOTES:
a. Dimensions are in inches.
b. Metric equivalents are in parentheses.
c. The input, output, and date of final electrical test shall be clearly marked on the tube.
d. The tube finish shall be cadium iridite on the gas switching portion (TR), and bronze iridite on the aluminum limiter section (output).
e. All dimensions specified herein are for qualification inspection.

FIGURE 1. Outline drawing of electron tube type DOD-008.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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